



A review of methods for gene regulatory networks reconstruction and analysis

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Accepted: 29 December 2024 / Published online: 29 May 2025
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Abstract

Gene regulatory networks (GRNs) are one of the most promising techniques for modeling and understanding biological processes. GRNs help to understand complex interactions involving genes. Regulator molecules control gene expression within these networks through interactions with each other and other substances in cells. They play a crucial role in deciphering regulatory relationships among genes and modeling changes in gene expression under various conditions. In recent years, there has been a surge in the generation of vast quantities and diverse varieties of gene expression data. This bulk production has created a demand for new data management methods. It is expected that integrative analysis of various types of gene information will provide a comprehensive overview for researching complex biological systems and processes. This review covers recent developments related to Gene Regulatory Networks and associated gene expression data from 2019 to 2023. Key topics are presented according to the data type used for network building, machine learning approaches, tools used in network inference, model optimization, and computational validation techniques. This paper presents an in-depth technical review of the techniques currently used for constructing GRNs, explores possibilities for future research, and aims to enhance the understanding of GRNs.

Keywords Gene regulatory networks · Predictive modeling · Gene expression data · Transcriptome

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1 Introduction

Gene–gene interactions can impact disease susceptibility and progression in exploring complex diseases. By delving into the interactions and influences among different genes within biological pathways, researchers can pinpoint crucial genetic elements that contribute to disease vulnerability. Grasping these interactions is essential for developing more precise diagnostic tools, tailored treatment approaches, and focused interventions for complex diseases. Network modeling has transformed our comprehension of biological mechanisms and has played a significant role in uncovering disease biomarkers. Extensive work has been focused on reconstructing different biochemical networks by utilizing functional genomic data produced through high-throughput techniques (AL-Bermamy and AL-Rashid 2021). Wang and Huang (2014) examines the statistical approaches employed in reconstructing gene regulatory networks using gene expression information.

The gene–gene interaction refers to one gene interacting with another gene; hence, it may affect the functional expression of another gene either directly or indirectly. Protein–protein interactions involve the interaction of proteins through binding to form functional protein complexes. The proteins, such as protein interactions, play important cellular roles, including signal transduction, metabolism processes, and environmental signaling. Regulatory gene interactions involve the participation of transcription factors or promoters that control the expression of genes, development, growth, and responses to environmental conditions. These interactions are necessary to understand regulatory interactions shaping complex biological processes. It is difficult to measure interactions within living cells. Measuring component abundances, such as mRNA levels, has become easier with technology advancements in the previous 2 decades, allowing for large-scale gene expression studies at lower prices. This trend has elicited a strong incentive for computationally modeling interaction structures that support gene expression patterns. Such interactions collectively form GRNs.

Systems Biology, through an interdisciplinary approach, has been very active in network reconstruction. GRNs can be defined as complex networks of interactions among genes, proteins, and several other molecules. At the heart of these networks are transcription factors: specialized proteins that interact with specific regions in DNA, controlling how different genes are activated or repressed. Through this means, the cell controls the production of proteins, a process commonly known as gene expression. Importantly, GRNs are not just constellations of genes toiling individually; these genes mutually inhibit or activate one another, establishing feedback loops through which cellular processes can be exquisitely fine-tuned. Combining these strongly interacting features enables the cell to detect and respond to including combining features enables the cell to detect and respond to internal signals, external stimuli, or both- and differentiate into a wide range of cell types, all while correctly executing specific functions. For all these reasons, understanding GRNs goes to the heart of how organisms grow and function and can become diseased. By unraveling these networks, researchers aim to find out how the different genes might coordinate with each other in complex biological processes and identify potential therapeutic targets.

Figure 1 depicts a cartoon illustration of direct regulation (gene 2 by gene 1) and combinatorial control via complex creation (gene 3 by genes 1–2). The (directed) network on the right depicts the system's abstracted structure. The complexity of relationships underpins gene expression; essentially, GRNs provide a glimpse into the elaborate communication system cells utilize to synchronize their actions, ultimately shaping the fundamental basis of

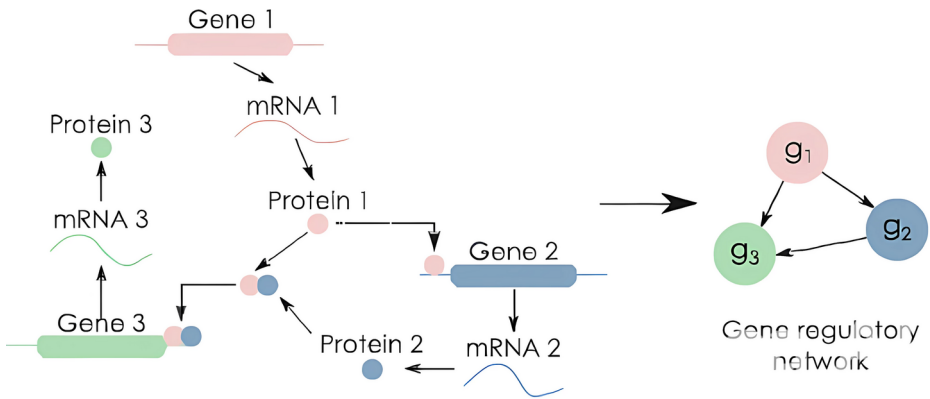


Fig. 1 Complex model describes the interaction between three genes (Huynh-Thu and Sanguinetti 2019)

life. Gene–gene interactions are paramount in shaping the intricacy of biological processes, especially in the realm of complex diseases and traits (Article 2008).

These interactions entail the impact of multiple genes on each other's expression, resulting in a wide array of phenotypic manifestations. Comprehending gene–gene interactions is imperative for deciphering the genetic foundation of complex diseases and traits, given their substantial contribution to individual variability. Specifically, we emphasize the advancements made and the challenges that remain in estimating gene interactions. With the rapid progression of technologies providing various large-scale genomic datasets, we also review techniques for integrating these additional data to enhance the accuracy of gene network inference. This paper examines methodologies utilized in reconstructing biological networks, specifically focusing on gene networks. The selection of the modeling approach is predominantly influenced by the nature and reliability of the available data, pertinent biological inquiries, and statistical and computational constraints. We collected different types of studies related to gene–gene interactions from 2019 to 2023. The method we focus on in this review is machine learning (ML). The significance of our review is that it provides insights into the evolving landscape of progress of all machine learning methods.

2 A brief overview of models to gene regulatory network

There have been numerous techniques for modeling and simulating gene regulatory networks. Models can be classified into gene regulatory network model classes (Karlach and Shamir 2008), including network topology, network control logic, control logic, and dynamic models. The outcomes from genome sequencing and other advanced technologies have offered insights into the complexity of molecular networks, revealing that they consist of tens of thousands of components, such as genes, with at least an equal number of connections among them. The challenge lies in understanding networks that encompass such vast numbers of elements. One approach to grasping these networks is developing models that simplify and represent complex systems like gene networks (Alrashid et al. 2022). These models are intentional simplifications designed to make the intricate details of reality more understandable.

This review discusses that simulations involve using a model to generate data, which can then be compared with experimental results. Conversely, reverse engineering consists of starting with data and attempting to construct a model that aligns with it through (semi-) automatic modeling techniques. In this approach, the model is crafted based on the data and validated through simulations compared against new experimental data. For instance, a gene expression dataset may be used to develop a gene network model that fits the data within a specific model class. Discrepancies between the simulated data from this model and new, uninvolved data can reveal shortcomings in the model. These discrepancies can be used to evaluate different models or to refine the existing model. However, the feasibility of simulations or reverse engineering hinges on sufficient quantitative data to depict the system's behavior. The emergence of extensive datasets, like microarray gene expression and genomic localization data, has fueled the search for sophisticated methods to model complex biological systems.

Topological models, essentially graphs depicting the links between elements, have been applied to various biological datasets, such as protein–protein interaction and coexpression networks. The structure of a gene network is portrayed as a graph, where nodes symbolize genes and edges or arcs denote the interactions among these genes. In the control logic model, the network's topology illustrates the dependencies between genes within the network. However, it does not reveal the regulatory significance of these dependencies, namely the control logic. Logical models provide a straightforward and effective approach, particularly useful when knowledge is limited, and have been effectively utilized to pinpoint specific regulatory interactions.

Dynamic models are widely recognized as conventional techniques for modeling gene networks. These models have been successfully used to identify specific regulatory interactions. They are seen as traditional approaches in gene network modeling, having been developed well before the onset of the contemporary genomic period. Their objective is to describe and often replicate the dynamic fluctuations within a system's state and to predict the network's reaction to various environmental changes and stimuli. It is also evident there is a contrast between models that depict network architecture and those that focus on network logic or dynamics—the former covers the entirety of the genome, while the latter typically focuses on simulating just a few genes. Additionally, high-throughput assays, like microarrays, offer temporal insights into transcriptional activities in time series studies (Alrashid et al. 2018).

Gene regulation networks can be simulated and modeled using various unique approaches. Apart from the Classifications mentioned above (topology models, control logic models, and dynamic models), they may encompass the following: Models of complicated networks that examine gene connections as intricate webs. Models of biological interactions focus on the relationships between chemicals, proteins, and genes. Machine learning models: employ methods to forecast the behavior of regulatory networks, such as random trees or neural networks. Models of chemo dynamics: examine variations in molecular concentrations throughout time. Models of molecular simulation are used to describe biological processes in detail. Depending on the research goals, each model type has advantages and is applied in various circumstances. These models are mentioned in the references Raza (2016) and Raza et al. (2014) and are complex network models that describe how genes are connected, how genes are connected, and how gene regulatory networks are examined. Biological interaction models Studies aim to understand how genes interact with each other and

affect biological processes, especially about cancer. Machine learning models where gene connections and regulatory networks are inferred using algorithms such as mutual information. Molecular simulation models and molecular dynamics through investigation of gene interactions can involve parts of the simulation such as understanding molecular dynamics through investigating gene interactions. However, the primary focus can involve parts of simulation, such as understanding molecular dynamics by investigating gene interactions. Therefore, it can be claimed that the research covers a variety of models, such as biological interactions, complex networks, and machine learning-based approaches.

3 Datasets

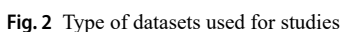
Predicting gene regulatory networks through machine learning techniques relies on dataset availability. Typically, such common Datasets fall into categories: Microarray data, which is widely available for various organisms and tissues; RNA-seq data, which advantages with more accurate quantification of gene expression (AL-Bermany and AL-Rashid 2021). Single-cell RNA-seq data is advantageous in revealing cell-type-specific gene expression patterns. The accuracy and reliability of GRN inference heavily depend on the quality of the underlying data. It is essential to carefully assess and address potential, noise, and technical variation sources. Table 1 provides a comprehensive overview of the diverse datasets used in all categories of common datasets, offering insights into the complex network of gene–gene interactions that govern various biological processes and diseases.

There are datasets particularly suitable for gene–gene interactions, such as time-series expression data (Bar-Joseph 2004), available on the DREAM Challenges website, which allow for studying changes in gene expression over time to infer dynamic gene regulatory networks (GRNs) and identify regulatory relationships based on temporal patterns. In

Table 1 This article reviews a summary of gene regulatory network datasets utilized by studies

Type	Dataset name example	Description	Resource
Microarray data	GEO (Gene. Expression Omnibus) Array Express	A public repository of microarray and high-throughput sequencing data public repository with a focus on functional genomics data	https://www.ncbi.nlm.nih.gov/geo/ https://www.ebi.ac.uk/arrayexpress/
RNA-seq data	TCGA (The Cancer Genome Atlas GTEx (Genotype-Tissue Expression) NCBI SRA (Sequence Read Archive)	A large-scale cancer genomics program with multi-omics data, including RNA-seq for various cancer types Aims to characterize genetic variation and gene expression across different human tissues A database that offers access to raw sequence data and contains RNA-Seq data from various studies	https://www.cancer.gov/tcga https://gtexportal.org/home/ https://www.ncbi.nlm.nih.gov/sra
Additional resources	dbGaP BioSamples Expression Atlas	Database of Genotypes and Phenotypes, linking genetic variation to phenotypes Provides information about biological samples used in research Discover up and down-regulated genes in numerous experimental conditions	Home-dbGaP-NCBI (nih.gov) Home-BioSample-NCBI (nih.gov) Home<Expression Atlas<EMBL-EBI

To gain valuable insights into tissue-specific regulatory mechanisms, the GTEx (Genotype-Tissue Expression) project aims to measure genetic variation and gene expression in several human tissues (Melé et al. 2015). In addition, the Human Cell Atlas initiative provides a complete reference map of all types of human cells and could contribute to understanding gene regulation in cellular contexts. A good example project is the DREAM project for Dialogue on Reverse Engineering Assessment and Methods. Marbach et al. (2016) provide a set of benchmark datasets on gene regulatory network inference methods by providing both simulated and real datasets with known network structures. ENCODE project—Encyclopedia of DNA Elements—identifies all functional elements in the human genome; this includes regulatory elements and transcription factor-binding sites, which can be used in data-driven gene regulatory network inference. An overview of the various studies analyzed within this manuscript and datasets are given in Fig. 2. This figure helps readers understand the broader context of the diversity of data and methods used in the studies in question.



4 Methodological for paper selection

The selection of articles included in this review was, thus, aimed at the inclusion of only high-quality research endeavors that were significant concerning the specific topic under consideration. Many of them were focused on genetic expression data. In addition, a considerable part of the reviewed studies aimed at realizing GRNs. We selected articles published between 2019 and 2023 to account for the most recent trends and developments in the use of gene expression data in GRNs. Selected articles were systematically grouped according to their specific research aims based on the methodology and diversity of the applied genomic data landscape in the area. Machine learning studies on gene regulatory networks deal with applications using computational algorithms and statistical models by the complex relationships in GRNs. Such relationships define how genes turn on or off at differential timings and environmental changes, which could result in diseases. To give a structured view of GRN usage on gene expression data, we classified these selected publications according to the research goals they targeted. This kind of systematic classification may help develop a full understanding of the genomic data environment and help comprehend the many methods applied in this field. The field consists of an array of inquiries and methodologies, and a few are depicted in Table 2.

The research articles evaluated here appeared in a wide range of forty-six periodicals. Figure 3 is a wise visual representation that captures the distribution of publications across many prestigious journals. Bioinformatics is the most prolific journal, contributing 46 publications under review. It is a leading platform for publishing innovative studies in computational biology and bioinformatics, especially those related to Gene Regulatory Networks in gene expression data. Each of these journals has a unique focus: Nature Communications showcases high-impact research from across the natural sciences; PLoS Computational Biology presents significant contributions that deepen our understanding of biological systems through computational methods; and Springer Nature, as an interdisciplinary, open-access journal, covers a broad spectrum of topics in engineering and technology. Journals like BMC Bioinformatics accounted for 22% of the reviewed publications. Of the reviewed

Table 2 Summary of studies employing gene–gene interactions in gene regulatory networks

Type studies	Aim	Studies
Network inference	Utilization approaches to infer interactions between genes, proteins, or other molecules in a biological system	Kc et al. (2019), Yu et al. (2021), Chen et al. (2019), Mao et al. (2023), Liu et al. (2020), Ma et al. (2020), Shachaf et al. (2023), Feng et al. (2023) and Zhang et al. (2023)
Gene expression analysis	Utilization approaches to identifying genes associated with diseases and cell type classification	Mostavi et al. (2020), Leng and Wu (2022), Liu et al. (2022), Bafna et al. (2023) and Xue et al. (2023)
Network topology analysis	Utilization approaches to identifying structural properties of the network (modules and subnetworks), Network dynamics, and time-series analysis	Devi and Singh (2020), Graafland and Gutiérrez (2022), Monti et al. (2022), Cui et al. (2022) and Jiang et al. (2022)
Predictive modeling	Utilization approaches to build mathematical or statistical models to predict a future event or outcome	Rothenberg (2019), Kotiang and Eslami (2020), Park et al. (2022), Zito et al. (2023), Li et al. (2023) and Jagodnik et al. (2023)
Integrative multi-omics data analysis	Utilization approaches to integrate various types of biological data to uncover complex gene regulatory relationships	Wani and Raza (2019), Paczkowska et al. (2020), Fofana et al. (2021), Wani and Raza (2021), Pažíňková and Popovici (2022), Jagtap et al. (2022) and Song et al. (2022)

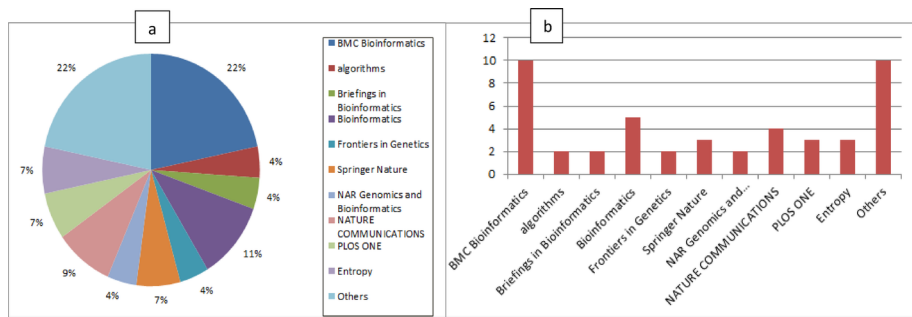


Fig. 3 a A proportional representation of papers in different journals; b journals with multiple publications

Table 3 Journal name and papers

Journal name	Num. papers	Percentage
<i>BMC Bioinformatics</i>	10	22
<i>Algorithms</i>	2	4
<i>Briefings in Bioinformatics</i>	2	4
<i>Bioinformatics</i>	5	11
<i>Frontiers in Genetics</i>	2	4
<i>Springer Nature</i>	3	7
<i>NAR Genomics and Bioinformatics</i>	2	4
<i>Nature Communications</i>	4	9
<i>PLoS One</i>	3	7
<i>Entropy</i>	3	7
Others	10	22

articles, bioinformatics represents 11% of articles, and algorithms, NATURE COMMUNICATIONS, PLOS ONE, and Entropy account for 7% each.

This published publication survey sheds light on the academic settings most receptive to GRN-based gene expression research. The diverse spectrum of journals reflects the interdisciplinary character of this study field, which combines bioinformatics, genomics, machine learning, and artificial intelligence. The distribution of articles among different journals, as indicated in Table 3, demonstrates the relevance and widespread acknowledgment of this unique research subject.

4.1 Recent studies of network inference

In recent years, inferring networks using computational methods has become a crucial aspect of the post-genomic era. To this end, the “Dialogue on Reverse Engineering Evaluation and Methods” challenge motivates researchers to develop advanced and efficient computational techniques. The rapid advancements in high-throughput technologies have led to numerous research initiatives generating vast quantities of gene expression data, which facilitate using computational methods to deduce gene regulatory networks from this information. There has been considerable progress in mapping out the structure of GRNs based on gene expression data. Initially, Boolean networks were favored for GRN inference, as outlined by Ruz and Goles (2013), with gene states indicated by Boolean variables and gene

interactions modeled by Boolean functions that define gene states influenced by other regulatory gene states. Nowadays, information-theoretic approaches are gaining prominence for reconstructing GRNs. Techniques based on mutual information, such as the relevance network (REL), context likelihood of relatedness (CLR), ARACNE, and minimum redundancy network, have been successfully applied to derive GRNs.

Link prediction can assist in supplementing missing data in real networks while allowing for a better understanding of network evolution. Linyuan and Zhou (2011) aims to infer missing linkages or anticipate novel interactions between non-connected network nodes. Numerous techniques for link prediction have been developed based on network topology. Some of these algorithms have been effectively employed in biological networks to solve a variety of biological and medical challenges, such as the prediction of gene regulatory relations (Barzel and Barabási 2013), protein–protein interactions (Kovács et al. 2019), lncRNA–protein interactions (Hu et al. 2017), drug–target interactions (Cheng et al. 2012), drug–drug interactions (Zhang et al. 2017), gene–disease associations (Lin et al. 2017), miRNA–disease associations (Chen et al. 2018). Most predictions are made in unipartite networks with only one set of nodes, while some are made in bipartite networks with two non-overlapping sets of nodes and only links connecting the nodes in distinct sets. Table 4 presents research on recent investigations undertaken on the subject.

Chen et al. (2019) presents D3GRN as a novel GRN inference technique that can infer gene regulatory networks from gene expression data. The approach aims to understand the problems encountered in complex relations between genes and how they regulate each other's expression. D3GRN has contributed by presenting a novel and effective approach for unraveling intricate regulatory relationships within biological systems. The complete GRN inference problem can be broken into small, manageable subproblems, each focusing on finding regulators for a single target gene. In that respect, finding network regulatory interactions becomes a manageable feature-selection problem. D3GRN uses ARNI to identify the regulators of individual target genes. ARNI is formulated using a functional decomposition framework, decomposing the expression of the target gene in terms of a sum of functions modeling potential regulators. D3GRN uses bootstrapping to establish confidence in the identified regulators. The algorithm resamples the data, runs ARNI on each resampled data set, and describes the behavior as a QV parameter to encode a multi-parameter space. The method uses a scoring scheme, giving each putative regulator a score according to how often and in what order it gets selected by ARNI throughout the bootstrapping runs.

Kc et al. (2019) developed deep learning structure, GNE, could embed both the gene expression data and gene interaction networks that allowed effective prediction of gene interactions. GNE aims to learn an embedding representation that integrates the known gene interactions and gene expression for predictions of gene interaction. To the best of our knowledge, the method regards the inference of gene interaction networks as a problem related to network embedding, wherein gene interactions are embedded into a network where genes and their interactions are realized. The GNE allows gene expression data to be incorporated with topological structure information of the gene interaction network for improved gene interaction prediction. In addition, the deep learning framework designed to predict gene interactions integrates information from gene interaction networks and gene expression data. GNE represents genes in a low-dimensional vector by using a network embedding approach. This embedding captures topological proximity, which are relationships based on network structure, and attribute proximity, which are relationships based on

Table 4 Studies in network inference

References	Contributions	Code availability
Chen et al. (2019)	Proposed a method D3GRN to network construction transforms each target gene's regulatory relationship into a functional decomposition problem, which is then solved using feature selection.	https://github.com/chenxofhit/D3GRN
Kc et al. (2019)	proposed a new deep learning framework for learning lower-dimensional representations of GI networks while preserving topological and attribute proximity, resulting in significantly better predictions.	https://github.com/ckekishan/GNE
Liu et al. (2020)	A new network inference technique, RWRNET, was suggested to enhance the precision of network inference.	Not available
Ma et al. (2020)	Introduced Non-linear Ordinary Differential Equation (ODE) models to simulate the dynamics of regulatory networks, offering the benefit of accurately modeling gene transcription and translation.	https://github.com/lab319/GRNs_nonlinear_ODEs
Yu et al. (2021)	Developed a novel method DCNTC to improve the GRNs' inference accuracy, achieving a high correlation measure of association between genes.	Not available
Mao et al. (2023)	Created a novel Predictor of potential interdependencies between genes in the network and inferred interdependencies, maintaining a lower computational complexity with datasets of varying scales	https://github.com/sdesignates/GNNLink
Shachaf et al. (2023)	Proposed a novel GRN inference algorithm (CMIA), achieving superior performance in GRN reconstruction compared to other methods; robustness and accuracy make it a valuable tool for GRN inference.	https://github.com/XiaoLab-JHU/GRN_Inference
Feng et al. (2023)	Proposed GRINCD , a novel framework for identifying key factors affecting cancer development, filling the gap of effective causal based.	Not available
Zhang et al. (2023)	The challenges of inferring gene regulatory networks using SCENIC + and CellOracle remain multidimensional and anything but trivial, which can be accomplished by continually improving computational methods and more understanding of genetic, and biological constructs.	https://github.com/Roy-lab/scMTNI

Bold indicates to emphasize the methodologies adopted in previous studies

gene expression similarity. It uses deep learning models with multiple hidden layers to learn the complex statistical relationships between gene network topology and gene expression data. Information such would permit a finer delineation of gene interactions than provided by methods employing only one type of information. This concatenation of topological and attribute embedding is placed as input into the deep learning model, which is helpful for better-capturing interactions between these two kinds of information than the approaches that use late fusion.

Liu et al. (2020) described a distinctive network inference technique that utilizes expression data for discovering complicated regulatory links between genes and discloses the mechanism of several diseases. A new network inference algorithm, RWRNET, is presented in this methodology. The approach encapsulates the local topology via three random walks and then fuses it with global topology using a Random Walk over Restart. Markov Blanket discovery algorithm was employed to care for solitary genes in the network. The model consists of three major constituents in RWR architecture that were enhanced using the method under discussion: Restart Probability determined as mean path length among genes repre-

senting the topological structure of a network, initial Probability Vector constructed based on gene expression, and functional modules. The hubs were detected based on high-expression genes treated as module initiators. Then, they utilized a new search algorithm, Roaming Network, by establishing an asymmetric mutual information ranking. The general idea behind this approach is to simplify regulatory relationships between genes by eliminating indirect connections. Since RWR is performed on a roaming network for each gene, we can get a stationary distribution of the probability of finding any other genes given by one specific gene. A final score was computed for every gene pair based on station distribution and transition probability; this adaptive threshold can detect regulatory relationships between genes. The IPC-MB (Markov Blanket) algorithm (Hu et al. 2017) is employed to infer the Markov blanket, representing a gene set directly regulating an individual gene. The following framework to optimize biological network structure with isolated genes is developing.

Ma et al. (2020) method can accurately reconstruct gene regulatory networks from time-series and steady-state measurements, performing the leading methods for artificial expression data and comparable real-life inference scenarios with faster runtimes. His methodology uses non-linear ordinary differential equations (ODEs) to describe the dynamical interaction among genes and XGBoost, a powerful machine learning method, to calculate the importance of possible regulatory connections. It is used to determine the significance score for each putative regulatory connection. This score is indicative of the regulatory crosstalk between genes. In this study, we proposed an efficient way of using non-linear ODE modeling with XGBoost to measure the importance of GRN inference. It is an indispensable tool that can be scaled with the method.

Liu et al. (2021) proposes the DCNTC (Distance Correlation and Network Topology Centrality) network to estimate gene regulatory networks from gene expression data. It addresses the non-linear dependencies and their sparsity in GRNs by combining distance correlation with network topology centrality. DCNTC uses distance correlation instead of mutual information to evaluate inter-gene dependency. The prime reason is that distance correlation is a fine measure for profile-based interpretation of nonlinear relationships without assuming the data to be distribution-free. The main reason is that it utilizes network centrality measures in its implementation, like degree centrality; hence, the importance of genes in the interconnect graph is captured. Identify hub genes that are highly connected and acting as key players in controlling the rest of the other genes. A distance correlation matrix was built for the pre-processed gene expression data with a threshold to trim off weak, insignificant links. The remaining pre-processed matrix of connections is then used to evaluate the degree of centrality of each gene. In the second step, the genes are ranked individually and awarded a secondary score that considers the centrality of the surrounding genes. A regulator is chosen as the gene with the maximum distance correlation from the higher-ranked gene list. This process is iterated for every gene to obtain the final GRN. The potential of DCNTC lies in capturing nonlinear dependencies and emphasizing critical links, making it one of the most useful tools in studying complicated biological networks and their regulating mechanisms.

Mao et al. (2023) introduced GNNLink, which applies graph neural networks to predict gene regulatory networks from scRNA-seq data. Gene regulatory networks are critical for decoding complex interactions. Consequently, due to the resulting regulatory mechanisms between genes, they provide extensive and profound insights into cell function and disease pathways. The GCN-based encoder allows us to capture gene dependencies effectively

and learn informative features. GNNLink first filters lowly expressed genes and focuses on teaching features of the genes that are highly variable. The GCN is utilized for learning gene features, while an aggregation of neighboring genes in the network captures the dependencies between the genes, allowing informative features to be encoded. The learned features of the genes are utilized to predict possible regulatory interactions between genes by completing a matrix. It makes it possible to predict probabilities with which any two genes are regulatory partners.

Shachaf et al. (2023) introduces a novel technique to improve gene regulatory network (GRN) inference through CMIA combined with the KSG-MI estimator. This method yields statistically significant improvements in precision-recall measures over previous best results, indicating that mutual information estimation and the kNN-based KSG algorithm are important for GRN inference methodologies. This paper has surmounted the shortcomings of traditional fixed binning methods by employing KNN-based estimation of mutual information, especially with the Kraskov–Stoicbauer–Grassberger (KSG) algorithm. Hence, more accurate and reliable GRN reconstructions have been achieved. More than that, this algorithm helps reduce the errors associated with estimates of mutual information and increases the effectiveness of GRN reconstruction; so, CMIA is suggested as the new promising inference algorithm.

Feng et al. (2023) presented GRINCD, one of the first-of-its-kind frameworks designed to infer Gene Regulatory Networks from gene expression data. In contrast to traditional methods depending on correlations alone, GRINCD employs causal discovery to identify the regulatory relationships between genes, giving a more accurate and biologically relevant image of underlying mechanisms. First, the GRINCD, in a first step, builds potential networks based on linear (Pearson) and non-linear (mutual information) relationships of genes by capturing the initial co-expression patterns. In the second step, exploring these complex interactions embedded in these networks, the advanced graph neural network algorithm called GraphSAGE is adopted. GraphSAGE aggregates information from all other genes to learn in-depth gene representations and bring light to the function of every gene in the network. GRINCD itself is based on the ANM to find the causal directions between pairs of genes, defined as the independence between the regression residuals and independent variables, giving the likely direction of influence. Such a causal discovery method is applied to both TF-TG and TF-TF pairs, offering a view of the whole regulatory network. Finally, GRINCD integrates results from its linear and nonlinear pipelines by a count voting mechanism; also, the capacity to handle large-scale datasets and detect both linear and nonlinear relationships makes the tool useful for understanding complex biological systems and diseases.

scMTNI (Single-cell Multi-Task Network Inference) in the article Zhang et al. (2023) introduces a new method to better understand regulatory networks by integrating many data sources. This method can be applied to many research projects, such as oncology studies, regenerative medicine studies, and regenerative medicine studies involving reprogramming and differentiation of cells. These results indicated that scMTNI outperforms the existing methods on network inference from single-cell data. The paper focuses on the recently developed structure of a framework known as scMTNI, which can analyze gene regulatory networks of a particular cell type using single-cell sequencing and chromatin accessibility. The study identified current methods lacking in integrating information from scRNA-seq and scATAC-seq toward the inference of gene regulatory networks. The scMTNI frame-

work was developed, integrating scRNA-seq and scATAC-seq data in a multi-task learning manner while considering cell lineage structure information. The study opened up new avenues of investigation by identifying known and unknown gene regulators implicated in cell transitions. To study to have a better understanding of the study's applicability, scMTNI has been applied to both real and simulated data to study cell differentiation and cell reprogramming.

Trends, challenges, and future directions in recent studies of network inference Recent studies have shown, in an archetypal manner, that considering Liu et al. (2020), Mao et al. (2023) and Liu et al. (2021), gene regulatory networks can be inferred from gene expression data through various methodologies and tools. These computational approaches aim to unravel complex gene interactions that are important to understanding biological processes. The dynamic network construction methods studied by Chen et al. (2019) are a step toward more sophisticated GRN inference strategies that integrate theoretical insight with computational tools to leverage our understanding of gene regulation to facilitate discoveries in systems biology. Furthermore, inference of GRNs from scRNA-seq data is a rather complex issue because of noise, sparsity, demand for large sample size, loss due to data transformation, and computational demand (see, for example, the work of Mao et al. (2023)). The computational challenges of the SCENIC+ and CellOracle methods in the inference of gene regulatory networks represent a multi-aspect, complex problem stated in the paper by Zhang et al. (2023). They will be solved only through continuous development in methods and a deeper understanding of genetic biology. Those challenges can only be met by state-of-the-art computational methodologies and technologies that increasingly enable gaining robust insights into gene regulatory mechanisms from complex biological data.

The research of Shachaf et al. (2023) and Ma et al. (2020) indicates a need for development beyond the current correlation-based approach to the inference of GRNs to account for the non-linear dynamics and to incorporate other data types. These studies present the emergent application of machine learning and deep learning, which are known to considerably improve the accuracy and efficiency of the GRN inference, as shown by Kc et al. (2019). The paper describes integrating multiple biological data sources to construct more detailed and precise gene networks. More importantly, most of these studies demonstrate the value of data integration and illustrate different ways of integrating gene interaction networks with gene expression data. They also consider scalability issues and methods to cope with large and complex data. Those are the challenges that future research has to be concerned with, given that a growing number of methodologies to infer GRNs combine machine-learning techniques with statistical methodologies. Using different graph neural network architectures or assembling methods, for example, the works of Mao et al. (2023), could greatly enhance model performance. In such regard, the DCNTC method proposed by Yu et al. (2021) is very promising for inferring gene regulatory networks. Analyzing time-varying gene regulatory networks or under different conditions gives information about the temporal factor in gene regulation. Eventually, it leads to developing algorithms that can capture dynamic network structure changes. Further research should be directed toward reasoning about deep learning models in Kc et al. (2019), to understand deep learning models and extract biological insights from the inferred networks. Dynamic network inference remains a challenge, and so does the future direction of all approaches.

4.2 Recent studies on predictive modeling

In the past, several studies have been carried out on applying predictive modeling in gene regulation. Patel and Bush (2021) developed a computational framework for integrating multi-omics data into a model of gene expression, which significantly improved predictive accuracy compared to models based on only cis-regulatory features. Stoof et al. (2019) have recently questioned the homogenous molecular assumption during bacterial gene regulation processes. In so doing, they propose a unified model that considers the role of molecular diffusion in network performance. Rothenberg (2019) discussed these challenges in extending models of gene regulatory networks to more diverse developmental systems, which include the effects of cellular history, dose dependence of transcription factor activities, and stochasticity of the transcriptional outputs. A probabilistic graphical model has been put forth by Kotiang and Eslami (2020) to enable a system-wide study of gene regulatory networks; their results have demonstrated a high correlation between observed states and system behavior in *Escherichia coli*. These, taken together, show the promise of predictive modeling for gene regulatory networks. Further work is required to remedy a number of the more troubling challenges.

Studies proved that graph convolutional networks for modeling gene expressions show promising results. Several such creative approaches have been proposed to improve the accuracy of gene expression prediction. For example, Li et al. (2023) effectively included the information of Gene Interaction Graph by using Neighbor Connection Neural Network (NCNN) and GraphReg. More importantly, the model developed by Jagodnik et al. (2023), the Heterogeneous Integrated Graph for Predicting Disease Genes (HetIG-PreDiG), takes an edge over the current methods due to the integration of gene-gene, gene-disease, and gene-tissue associations. Success rates are high in predicting novel disease genes.

Such methodologies can provide a more complete view of the complicated cross-talk between genes under different biological processes and diseases. Moreover, the contemporary focus critically relies on integrating diverse data types related to health, including genetic, environmental, and behavioral factors, to build holistic models for disease risk prediction—for example, as reviewed by Dias and Torkamani (2019). Additionally, artificial intelligence and machine learning techniques are increasingly used in refining genotype-to-phenotype predictions, which are important in understanding complex diseases' gene–gene interactions. Another key area of research pertains to the specificity of the interactions, especially with attempts to correct for non-specific hybridization in microarray data. Techniques have been developed to filter out data according to recognized specificity effectors to enhance the accuracy of predictive models in focusing on gene–gene interaction, as done by Koltai and Weingarten-Baror (2008). A summary of recent studies conducted in this field is given in Table 5.

Rothenberg (2019) proposed a graph convolutional neural network framework to predict gene expression by integrating spatial genomic organization and local regulatory factors, addressing the importance of long-range interactions in gene regulation. This paper proposes GC-MERGE, a novel framework for predicting gene expression by integrating local histone modifications and long-range spatial interactions using graph convolutional networks (GCNs). The GNNExplainer method is adapted and incorporated into the framework to interpret the model's predictions. It identifies the most important nodes (genomic regions) and features (histone modifications) contributing to a specific gene's predicted expression.

Table 5 Studies predictive modeling

References	Contributions	Code availability
Rothenberg (2019)	Introduces GC-MERGE, an innovative framework based on graph convolutional networks (GCNs) for gene prediction. This new approach successfully addresses the shortcomings of previous models that struggled to capture the three-dimensional structure of the genome.	https://github.com/rsinghlab/GC-MERGE
Kotiang and Eslami (2020)	Development of a computational framework (FGN) combining probabilistic graphical modeling and statistical estimation to explore gene regulatory networks, refining computational methods.	https://github.com/kotiang54/PgmGRNs
Shu et al. (2021)	Propose DeepSEM to achieve realistic scRNA-seq profiles and greater GRN consistency.	https://github.com/HantaoShu/DeepSEM
Park et al. (2022)	A unique statistical estimation of gene networks sensitive to stomach cancer treatments has been proposed.	Not available
Zito and Cutello (2023)	Create a complex network to improve gene expression simulation and gene regulatory network inference considerably, providing artificial and empirical evaluation benchmarks	Not available
Li et al. (2023)	Suggested Neighbor Connection Neural Network (NCNN) utilizes gene interaction graphs to enhance prediction accuracy	https://github.com/Xuanyu-Li/NCNN
Jagodnik et al. (2023)	Development of the Gene-Disease Prioritization Score (GDPS) measures the association level between any gene and disease by integrating Network Structure and Biological Similarity. This approach addresses the issue of arbitrarily choosing non-associated gene and disease nodes as negative training samples.	https://github.com/bartala/disease_gene

This study employed a GraphSAGE GCN model to learn node embeddings that effectively capture local histone modification patterns and long-range interactions within the graph's structure. The model performs several layers of graph convolutions, each aggregating information from the current node and all neighbors with their features. A fully connected network, in turn, uses these node embeddings to classify between active and inactive nodes using a Softmax output layer. In the regression task, embeddings for predicting each gene's continuous expression levels. It was shown in this paper that GCNs are effective for the integration of multimodal data and predicting gene expression, thereby gaining insight into the biological mechanisms driving gene regulation.

Developed a computational framework to integrate probabilistic graphical modeling with high-throughput biological data to study gene regulatory networks. Thus, this research focuses on the role of probabilistic models in comprehending complex biological systems. The goal is to quantify how well experimentally verified GRNs correspond to observed expression profiles across diverse conditions. This type of GRN uses the factor graph representation, which allows for dealing with the complex structures of the network and includes uncertainty in regulatory relationships. The FGN model based on experimental data uses the LBP algorithm to make inferences. The LBP algorithm is applied to make inferences in the FGN model and approximate marginal posterior distributions over gene states based on experimental data. The process combines a priori knowledge of GRNs with high-throughput gene expression data to improve inference accuracy and dig deeper into system-wide behavior. To handle the continuous data for gene expression, discretization for each gene is done

in a gene-specific manner using Gaussian mixture models to handle the noise arising from experimental measurements in the data. High prediction accuracy was still preserved when the model was modified to simulate a gene knockout experiment, indicating that the model can learn regulatory interactions under any condition.

In this paper, the authors describe the development of DeepSEM, a deep generative model presented in this study to handle issues while evaluating scRNA-seq data. DeepSEM seeks to identify complicated regulatory relations between genes, and how genes affect each other's expression levels is analyzed through it. Biologically Meaningful Representation of ScRNA-seq Data: The model learns a low-dimensional representation of data that captures the underlying biological diversity between cells. DeepSEM provides a generalization of the standard SEM implementation, which is applied mainly to causal inference, making room for including all nonlinear neural networks and thereby enabling the model to capture complex and nonlinear interactions among genes. We use a specific neural network layer, the GRN layer, to model gene regulatory interactions explicitly. This layer will serve as a biological constraint, shrink the model's parameter space, and make it more interpretable. DeepSEM can realize low-dimensional representation learning of the scRNA-seq data and simultaneously estimate the structure of the gene regulatory network based on the Beta-VAE framework. In its ability to capture complex gene interactions with accuracy in biologically meaningful ways, DeepSEM emerges as very important for single-cell research. Consequently, DeepSEM is a powerful and versatile tool that can be used for scRNA-seq analysis and inference in GRNs.

The strategy by Park et al. (2022) in predictive gene network estimation in gastric cancer drug response prediction successfully identifies drug response markers and molecular interactions supported by previous studies. This paper's methodology involves developing the predictive network method to estimate gene networks and predict specific biological characteristics simultaneously, incorporating network biology and minimizing errors in network estimation and prediction. The technique adapts network biology in the prediction model, minimizing network estimation and prediction errors. It performs very well in feature selection and prediction accuracy, particularly in identifying markers that govern drug response differences and other regulatory systems between drug-sensitive and resistant cell lines. According to the method, the drug-resistance ability of some specific gene suppressors and inducers can be manipulated, reducing drug resistance against cancer cell lines. The latter approach is based on known biological facts; for instance, it assumes that connected genes are in the same network and that hub genes play critical roles. This guarantees that interpretable and biologically relevant results will be obtained. This technique promotes the similarity of coefficients of the connected genes in the prediction model and allows for selecting important hub genes and their connection. This feature makes this method particularly relevant in translating the derived findings into clinical applications because of interpretability and biological relevance. Monte Carlo simulations showed that the Predictive Network significantly outperformed existing prediction accuracy and feature selection methods. The results indicated a large difference in regulatory systems between sensitive and resistant cell lines, establishing the importance of concrete molecular interactions in drug response.

Zito and Cutello (2023) presents Gene simulation and the inference of gene regulatory networks while using machine learning techniques, which have been major fields of interest in the past few years. The modeling being agent-based and incorporating temporal dependencies provides the model with the capability of capturing realistic gene interactions and

furnishing valuable insights for biological research. The specification is of a framework called the Artificial Environmental Setting (AES) operating in terms of the individual agents used to represent genes. Each agent is modeled as a neural network that predicts the expression level of its corresponding gene based on the expression levels of other genes at previous time steps. Each gene represents an agent, allowing for individual and dynamic behavior. In the method involving ordinary differential equations, like in similar approaches, separate equations are developed for each gene to analyze its dynamic behavior. Different types of neural networks (FCNN, CNN, RNN) are employed and selected through a metaheuristic optimization algorithm that can best capture the gene expression dynamics for neural network architectures. The agents take into account the history of gene expression through a state matrix, capturing the temporal dependencies and trends in the data; the model can infer the regulatory relationships and identify potential regulators by analysis perturbing the expression level of specific genes and outputs a regulatory matrix capturing the influence between genes and a probability matrix indicating the likelihood of regulatory relationships.

Discussed by the authors Li et al. (2023) presented a novel neural network dubbed Neighbor Connection Neural Network (NCNN) to use the gene interaction graph information, which incorporates graph information in a better way. The gene interaction graph is first sparsified by removing weak connections between genes based on a threshold. Helps capture the local patterns within the graph more effectively. The model utilized one layer of neighbor connection to avoid deteriorating gene expression information. Considering the weaker gene interaction signal than expression levels, this layer propagates information from a node's neighbors (genes with connections) to the node itself. Each connection is scaled by a unique trainable parameter, allowing the model to learn the importance of each interaction. The neighbor connection layer is flattened into a vector and fed into a traditional fully connected neural network (MLP) for further processing and prediction of learned representations. NCNN is in leveraging gene interaction information for gene expression prediction.

Jagodnik et al. (2023) HetIG-PreDiG utilized a heterogeneous graph approach, integrating various biological datasets, to predict potential gene-disease associations more accurately than existing methods. A heterogeneous graph is built based on a combined gene-disease graph created using known gene-disease associations, and a gene-gene graph is made based on gene-tissue interaction data, connecting genes found in the same tissue. Some gene-disease associations are randomly selected and removed from the graph for the model to learn from representing known associations. Unconnected gene-disease pairs with lower Gene-Disease Prioritization Scores (GDPS) are chosen to address the issue of potentially unknown disease genes in negative samples. The node2vec algorithm is applied to the graph to learn low-dimensional representations (embedding) of genes and diseases, capturing the network structure and node relationships. Gene-disease prioritization Score (GDPS-score reflects the degree of association between a gene and a disease) provides information beyond the network structure, considering biological similarity even for unconnected genes and diseases. The aggregated node embedding and GDPS are combined into a single feature vector, and a logistic regression classifier is trained on this feature vector to predict whether or not a gene-disease pair will form a link (association).

Trends, challenges, and future directions in recent studies of predictive modeling In the field of the Deep learning revolution, deep learning models, such as D-GEX, NCNN, and various CNN architectures, are demonstrating superior performance in capturing the intri-

cate non-linear relationships within gene expression data. It allows for more accurate predictions than traditional methods such as linear regression. There remains a need to include gene interaction networks to take in a broader perspective regarding the regulation of genes. NCNN and GC-MERGE, work by Li et al. (2023) and Rothenberg (2019) respectively, show graph-based techniques that potentially support exploiting this supplementary network information for enhanced prediction accuracy.

Building more comprehensive and robust models is already required to be based on combining various data modalities like gene expression, protein–protein interactions, gene ontology, and chromatin conformation data. This line only proves that combining multi-omics data, HetIG-PreDiG, and GC-MERGE (Jagodnik et al. 2023) can project strength toward holistic comprehension of the biological system at hand and better prediction performance. Apart from model interpretability, some techniques, like GNN Explainer within GC-MERGE and the analysis of regulatory matrices in HetIG-PreDiG, also shed light on how exactly their predictions are biologically warranted, engendering trust and potentially leading to deeper biological insight. The training process for deep learning models can often be computationally intensive and time-consuming, especially with large and complex datasets. Hence, more efficient algorithms and high-performance computing should be developed to alleviate these limitations. Models trained on specific cell lines or datasets alone might not generalize easily across different contexts. There is, therefore, the need to build strong, more adaptable models by varying cell types, experimental conditions, and biological systems.

Suppose some efficient, scalable algorithms can be implemented well, including novel GCN structures and further optimization strategies of large-scale biological data analysis. In that case, the future of gene expression prediction and disease gene identification will be full of promise, with exciting avenues to explore. Information derived from multiple modalities—proteomics and metabolomics data, single-cell sequencing, imaging, and the electronic health record—implicates a deep understanding of biological systems and models built with enhanced accuracy. More robust and informative approaches to AI model evaluation—for instance, in attention mechanisms or possibly alternative explanation strategies—are imperative in making strides toward greater insight into biology and more trust in therapeutic applications. With the resolution of these challenges and the pursuit of future directions, AI has great potential to change our notion toward gene regulation, mechanisms of diseases, and personalized medicine and eventually deliver superior diagnostics, treatments, and health outcomes for all.

4.3 Recent studies of gene expression analysis

Gene expression analysis for gene–gene interactions is crucial in understanding biological processes and disease mechanisms (Hadi et al. 2022). Gene expression analysis in GRNs involves studying gene interactions to understand biological processes. Various methods have been developed to infer GRNs, such as differential network inference algorithms (Leng et al. 2022), deep learning models incorporating gene interaction graph information (Xue et al. 2023), and tools like CLARIFY that simultaneously infer gene regulatory networks and cell–cell interactions using spatial gene expression data (Bafna et al. 2023). These approaches collectively contribute to a deeper understanding of gene interactions within regulatory networks, aiding in uncovering causative mechanisms in various biological processes.

Reconstructing gene regulatory networks from gene expression data can effectively identify regulatory connections between genes and offer more insight into biological control mechanisms. Table 6 lists recent studies conducted in this area.

This study Mostavi et al. (2020) aimed to develop and evaluate convolutional neural network (CNN) models based on gene expression data for accurate cancer-type prediction. A key objective was addressing the potential bias introduced by tissue-of-origin effects and interpreting the models to identify biologically relevant cancer marker genes. Three CNN architectures were designed: the 1D-CNN model takes factorized gene expression data as input and utilizes one-dimensional convolutional kernels to capture global features, and the 2D-Vanilla-CNN model reshapes the gene expression data into a 2D matrix and employs 2D convolutional kernels for feature extraction, similar to image recognition tasks. The third model, 2D-Hybrid-CNN, is inspired by ResNet architecture. It utilizes two parallel 1D convolutional layers (vertical and horizontal) applied to the 2D input matrix to capture local and global features. The 1D-CNN model was further applied to predict breast cancer subtypes using the PAM50 classification.

This study Leng et al. (2022) employs Interaction-Based Transcriptome Analysis (IBTA) through the Co-hub Differential Network (CDN) inference algorithm and the pivot APC2 metric to uncover underlying mechanisms in diseases such as colorectal cancer and COVID-19. The CDN algorithm is designed to deduce differential networks by analyzing gene interactions specific to IBTA. It promotes the identification of hub genes, which interact extensively within a biological network. This is achieved using gene weights derived from Differential Expression Analysis (DEA) and applying the CDN method to deduce networks for further examination. CDN computes the pAPC2 for each relevant interaction, generating matrices for two groups and aiding the differential network deduction. s algorithm uses Gaussian graphical models to assess conditional independence among genes and deduces networks based on a multivariate Gaussian distribution assumption. Through this

Table 6 Gene expression analysis

References	Contributions	Code availability
Mostavi et al. (2020)	Proposed CNN models offer high accuracy, interpretability, and the ability to identify biologically relevant cancer markers, improved diagnosis, prognosis, and personalized treatment strategies	https://github.com/chenlabgcrcri/CancerTypePrediction
Leng and Wu (2022)	Developed an algorithm known as the Co-hub Differential Network Inference (CDN) approach, which has shown significant analytical capabilities in case studies	https://github.com/Wu-Lab/IBTA
Liu et al. (2022)	An innovative method (JDINAC) was used to develop gene interaction networks based on SNP data, going beyond the typical focus on single-gene effects and providing a more comprehensive perspective of the genetic landscape of breast cancer	Not available
Bafna et al. (2023)	CLARIFY is proposed as a multi-level graph autoencoder (GAE) for refining intracellular and extracellular contact networks. CLARIFY was an effective tool for researchers to better understand these complicated biological processes and their spatial relationships	https://github.com/MihirBafna/CLARIFY
Xue et al. (2023)	Constructed a novel co-expression network (WGCNA) in psoriatic lesions to successfully treat moderate and severe psoriasis and prevent recurrences	Not available

Bold indicates to emphasize the methodologies adopted in previous studies

approach, the CDN algorithm effectively detects variations in gene interactions across different groups, enhancing the study of interaction-based transcriptomes.

This study Liu et al. (2022) aimed to explore the genetic mechanisms behind breast cancer (BC), specifically focusing on gene–gene interactions rather than single-gene effects. The JDINAC method was applied to construct gene interaction networks, identifying differences between BC patients and controls. This analysis was with no adjustment, adjusting for BMI, adjusting for menopausal status, and adjusting for both. Logistic regression was used to evaluate the association between hub gene SNPs (identified from the network analysis) and BC risk. GO and KEGG pathway enrichment analyses were performed to understand the biological functions of the interacting genes. This study aimed to understand better the complex genetic mechanisms contributing to BC by exploring gene interaction networks.

Bafna et al. (2023) the main goal of CLARIFY is to improve the understanding of cell–cell interactions (CCIs) and GRNs within a tissue by jointly refining both types of networks using spatial transcriptomics data. Existing methods typically focus on inferring either CCIs or GRNs separately. Still, CLARIFY aims to bridge this gap by considering the interconnected nature of these processes within the spatial context of a tissue. CLARIFY utilized a novel multi-level graph autoencoder (GAE) framework that incorporates information from both cell-level (spatial proximity) and gene-level (gene regulatory networks) interactions. The cell-level graph represents individual cells as nodes and connects them based on spatial proximity. A gene-level graph treats each cell as a subgraph containing its underlying GRN and establishes edges between genes of neighboring cells based on known ligand–receptor interactions. Two parallel GNN encoders learn latent representations for cell-level and gene-level features, capturing information about gene expression, spatial context, and gene regulatory interactions. Cell-level and gene-level embedding are aggregated to create a comprehensive representation that encapsulates the complex interplay between CCIs and GRNs. The combined embedding is used to refine the CCIs' reconstructed adjacency matrix for both cells and the genes simultaneously.

Xue et al. (2023) investigates the potential of oxymatrine, a natural compound, as a treatment for psoriasis, a chronic inflammatory skin disease. The study uses techniques, including full-length transcriptome analysis and weighted gene co-expression network analysis (WGCNA), to understand how oxymatrine affects gene expression and improves psoriasis symptoms. The study identified several gene modules associated with PASI score improvement, with the “light-green” module showing the strongest correlation. Further analysis of this module highlighted key pathways involved in psoriasis. The study identified five hub genes (CNFN, S100A8, SPRR2A, SPRR2D, and SPRR2E) within the light-green module, all related to the EDC, which plays a crucial role in epidermal differentiation and barrier function. Oxymatrine treatment appeared to normalize the expression of these genes, suggesting a potential mechanism for its therapeutic effect.

Trends, challenges, and future directions of gene expression analysis studies There is a growing recognition that analyzing interactions between genes provides a more comprehensive understanding of biological processes and disease mechanisms. Despite the advancements highlighted in this paper, several challenges remain. The five research papers you provided collectively highlight the increasing power and potential of combining machine learning, network analysis, and spatial transcriptomics to unravel the complex interplay between genes, cells, and diseases. Papers Leng and Wu (2022), Liu et al. (2022), Bafna et al. (2023) and Xue et al. (2023) all underscore the importance of understanding how

genes work together and focus solely on differential expression. IBTA Leng and Wu (2022) emphasizes analyzing transcriptome data at the interaction level using pAPC2, revealing crucial insights missed by traditional gene-based approaches. WGCNA Liu et al. (2022) identifies key gene modules and hub genes associated with oxymatrine's therapeutic effect in psoriasis, suggesting coordinated network responses. CLARIFY Bafna et al. (2023) showcases the value of jointly refining cell–cell interactions and cell-specific GRNs, demonstrating their interconnectedness and spatial dependence.

JDINAC Xue et al. (2023) constructs a gene interaction network in breast cancer, revealing hub genes and highlighting the potential of network analysis in understanding complex diseases. Integrating spatial information through spatial transcriptomics is revolutionizing our understanding of biological systems. Papers Bafna et al. (2023) and Mostavi et al. (2020) explicitly leverage spatial data to improve their models' accuracy and biological relevance. Machine learning and network analysis are becoming essential tools for deciphering complex biological data and extracting meaningful insights. All five papers employ different techniques to various extents, as in the above table.

Despite the advances noted in these publications, significant obstacles remain, including the need for larger and more diverse datasets, particularly in spatial transcriptomics, to assure model generalizability and robustness. Developing tools to analyze large machine learning models is critical for understanding the biological principles that drive predictions and converting results into clinical applications. Combining several omics data types (genomics, transcriptomics, proteomics, and so on) can provide a complete picture of biological systems and disease processes; there are multiple exciting opportunities for future study, including the development of more sophisticated and interpretable models capable of dealing with the complexity and variability of biological data while providing insights into the underlying mechanisms. Including prior knowledge and biological constraints is essential to direct the construction of models and guarantee the biological relevance of their predictions. Furthermore, designing experimental validations is crucial to verify anticipated interactions and biomarkers. By addressing these challenges and pursuing these lines of inquiry, the convergence of machine learning, network analysis, and spatial transcriptomics can revolutionize our understanding of biology and medicine in ways that lead to tailored medicines and better illness management.

4.4 Integrative multi-omics data analysis

The interaction of genes may involve integrating multiple layers of omics information, such as epigenetics, transcriptomics, and proteomics, in conjunction with metabolomics. These data sets could project a comprehensive picture of the activities occurring at the molecular level and underpin biological processes like disease progression, aging, and obesity. Accurate and full mapping from this multi-omics data requires advanced computational means, such as deep learning algorithms and Bayesian networks. Such methods supply the identification of critical genes and their regulatory pathways while giving insight into complex connections between many molecular components. Multi-omics integrative analysis is, therefore, a very strong approach for in-depth investigation into complex biological systems and opens the window toward personalized medicine that allows ante-derived customized treatment interventions. Table 7 shows current investigations performed in this area.

Table 7 Studies in multi-omics data analysis

References	Contributions	Code availability
Wani and Raza (2019)	It first recognized what was lacking in the existing methods of integrating scRNA-seq and scATAC-seq data in the inference of gene regulatory networks. Following that, a framework named scMTNI has been developed to integrate the data of cell lineage structure representation with scRNA-seq and scATAC-seq through multi-task learning	https://github.com/waninisar/MKL-GRNI
Paczkowska et al. (2020)	ActivePathways was an innovation in integrative pathway enrichment analysis for multi-omics data aimed at improved systems-level understanding of biological processes and disease mechanisms	https://github.com/reimandlab/ActivePathway
Wani and Raza (2021)	This study opened new possibilities for research by identifying both known and unknown gene regulators that may be implicated in cell transitions. ScMTNI was applied to stem cell differentiation and reprogramming in real and simulated data	Not available
Fofana et al. (2021)	Introduces a novel framework for integrating assumption testing and network information into gene expression analysis	Not available
Pačínková and Popovici (2022)	Using IntOMICS, a unique approach to biological knowledge discovery, resulted in higher accuracy than a previously described algorithm for gene expression data	https://gitlab.ics.muni.cz/bias/intomics
Jagtap et al. (2022)	Introduced (BRANEnet) network integration with multilayered heterogeneous graph embedding and demonstrated its effectiveness compared to existing techniques	https://github.com/Surabhivj/BRANEnet
Song et al. (2022)	The proposed (SMGR) innovative method for integrated analysis of scRNA-seq and scATAC-seq data reveals inherent molecular foundations, which improves knowledge of underlying mechanisms	https://github.com/QSong-github/SMGR

In the paper by Wani and Raza (2019) proposed a new method for Gene Regulatory Network (GRN) inference: Integrative Matrix Tri-Factorization for GRN inference (iMTF-GRN). iMTF-GRN utilizes multiple biological datasets (e.g., gene expression profiles, protein–protein interactions (PPI), and Gene Ontology (GO) annotations) to formulate an integrative relational framework. Consequently, fused data matrices are separated into three non-negative factors using a non-negative matrix tri-factorization technique to support interpretability. Consequently, fused data matrices are separated into three non-negative factors using a Non-negative Matrix Tri-Factorization technique to support interpretability. The general objective of iMTF-GRN is to extract latent structures relevant to known gene interactions while reconstructing the input relation matrices. We use the NMTF objective function, which minimizes the reconstruction error between the original relation matrix and its low-rank reconstruction. The target matrix to be finally reconstructed by the program should show potential regulative relationships between transcriptional factors (TFs) and their target genes. This implementation uses random number initialization and multiplicative update rules for iterative updates to the factor matrices until convergence criteria are met. The program selects, for predicting novel interactions, combinations of genes and regulators scoring above the threshold set by the user when compared against known regulators.

This paper by Paczkowska et al. (2020) employs Interaction-Based Transcriptome Analysis (IBTA) using the Co-hub Differential Network (CDN) algorithm and the pivot APC2 metric to uncover underlying mechanisms in diseases such as colorectal cancer and COVID-19. The CDN algorithm is designed to deduce differential networks by analyzing gene interactions specific to IBTA. It promotes the identification of hub genes, which are central in realistic biological networks due to their extensive interactions with numerous other genes. The algorithm calculates gene weights using DEA and employs the CDN method to derive networks for further analysis. CDN assesses the pAPC2 for each relevant interaction to generate matrices for two distinct groups, aiding in the differential network inference. Using Gaussian graphical models, the CDN algorithm measures the conditional independence among genes and deduces differential networks under a multivariate Gaussian distribution. This approach allows the CDN algorithm to detect variations in gene interactions across different groups, significantly enhancing interaction-based transcriptome analysis.

According to Wani and Raza (2021), the parallel execution framework of MKL-GRNI allows for the inference of gene regulatory networks based on a wide array of biological data, including DNA methylation profiles characteristic of gene expression and protein-protein interactions. In essence, the inference process involves gathering a number of datasets that come in relation with gene regulation, and further, according to their nature, using the appropriate kernel functions to transform these into kernel matrices. The GRN inference problem is treated separately for each TF as a classification model, framing it as a binary classification. The model was trained by 10-fold cross-validation on known regulations to make this computation robust. Application deployed a multiprocessing Python module to split the inference task into multiple subproblems corresponding to each TF and conduct each subproblem in parallel on different processors to achieve full computing. The separate outputs of the sub-processes are collected into a shared queue and combined to create the final output matrix representing the inferred GRN.

Fofana et al. (2021) presents a new statistical method in the analysis of gene expression data under the name Hybrid-Network. Two important aspects must be considered: statistical assumptions and gene network information to increase the accuracy and power of the analyses. This procedure combines p-values from various statistical tests depending on whether the assumptions are met. This allows a more general and robust analysis than possible with a single, potentially inappropriate test. It uses gene network information by assuming that all neighbor genes in the network have similar prior probabilities. This reflects the biological reality that genes with related biological functions or pathways often have correlated expression patterns. This latent variable model considers the probability of a gene being expressed and incorporates both the network's hybrid p-values and spatial information. Markov Chain Monte Carlo simulations are performed to estimate model parameters and posterior probabilities, enabling inferences regarding gene expression and network relations.

This paper Pačínková and Popovici (2022) introduces IntOMICS, a novel Bayesian network-based framework integrating multi-omics data with prior biological knowledge toward gene regulatory network inference. Its newness is in adding to already existing knowledge empirical biological knowledge learned directly from data for a full understanding of the regulatory relationships. Integration of multi-omics data, such as gene expression, DNA methylation, and copy number variation, must be done if complex biological processes are to be unraveled and personalized medicine methods are to be developed. The IntOMICS framework used Bayesian networks to model probabilistic associations between genes and

other molecular features. Additionally, it used adaptive MCMC and Markov blanket resampling to make efficient inferences of the network and avoid data discretization, hence avoiding a loss of information and sometimes capturing linear relationships between features.

In this paper Jagtap et al. (2022), BRANenet is a novel method for integrating multi-omics data by embedding information from multilayer heterogeneous networks into lower-dimensional space. This methodology exploits graph representation learning to model relationships between various types of biomolecules, such as genes, transcription factors, and metabolites. Capture and represent relationships between biomolecules from different omics layers in a shared embedding space. The methods used in this paper are Intra-omics networks (e.g., gene co-expression) are built based on pairwise correlations within each omics data type, and Inter-omics networks (e.g., TF-target interactions) are constructed using a priori biological knowledge from databases like YEASTRACT and Yeast Pathways. These networks are combined to form a multilayer network represented by a supra-adjacency matrix. A random walk-based Positive Pointwise Mutual Information (PPMI) matrix is computed, capturing the context and relationships between nodes in the multilayer network. The PPMI matrix is factorized using Singular Value Decomposition (SVD) to obtain low-dimensional embedding vectors for each node. Gain insights into the coordinated regulation of gene expression and cellular metabolism by analyzing the integrated network to understand complex biological processes.

Song et al. (2022) introduces SMGR, a robust and adaptable method for analyzing single-cell multi-omics data, facilitating a more nuanced understanding of gene regulation across various biological scenarios. SMGR is a pioneering technique for handling single-cell multi-omics data, including scRNA-seq and scATAC-seq data from diverse samples. Its primary aim is to discern co-regulatory programs, clusters of genes with similar expression patterns, and chromatin accessibility, indicating potential co-regulation. For more straightforward comparisons with scRNA-seq data, scATAC-seq peak counts are transformed into a gene-based activity matrix. Depending on the specific data traits, a Negative Binomial (NB) or Zero-Inflated Negative Binomial (ZINB) distribution is selected to model the data. A generalized linear model detects latent gene and peak representations while adjusting for technical noise and batch effects. The parameters for the chosen distribution (NB or ZINB) are determined through maximum likelihood estimation using the iteratively reweighted least squares (IWLS) method. These latent representations are refined via Markov Chain Monte Carlo (MCMC) simulations, employing a Metropolis-Hastings sampler. The refined data are clustered using k-means, with the optimal cluster number identified by the Bayesian information criterion (BIC). Each cluster delineates a co-regulatory program. Functional enrichment analyses of genes within these programs are performed using tools such as GO and Reactome. Additionally, tools like GENIE3 are used to predict upstream regulatory transcription factors, shedding light on the regulatory mechanisms involved.

Trends, challenges and future directions in recent studies of multi-omics data analysis
In the Integration of Multi-Omics Data, combining various omics data types like scRNA-seq, scATAC-seq, CNV, DNA methylation, and others provides a more comprehensive understanding of gene regulation. This trend is exemplified by all the presented methods, which leverage the complementary information from different omics layers to reveal a more complete picture of regulatory mechanisms. Integrating biological prior knowledge from databases like KEGG and ENCODE improves network inference's accuracy and biological relevance. **IntOMICS** and **BRANenet** demonstrate the effectiveness of this approach,

while SMGR leverages prior knowledge in downstream analysis of co-regulatory programs. The increasing availability of spatial transcriptomics data (as shown in **ActivePathways**) offers opportunities to integrate spatial context into future multi-omics analyses. Combining data from different platforms and technologies requires careful preprocessing and normalization to account for technical variations and batch effects. This remains a significant challenge, as seen in the efforts of **SMGR** and **BRANenet** to address these issues. Understanding the biological mechanisms underlying complex models is crucial for translating findings into actionable insights. While methods like **ActivePathways** provide evidence for enriched pathways, further investigation is needed to understand the specific regulatory interactions within those pathways. In particular, large-scale multi-omics data sets analysis can become computationally intensive for methods like IntOMICS involving Bayesian network inference and MCMC simulations. More efficient algorithms need to be developed to scale up such analyses. More complex model building: The community should seek to build more advanced machine learning strategies to capture nonlinear correlations and complex interactions across omics layers. Expanding the scope of analysis to more extensive omics data indexes the cellular process and regulatory mechanisms more comprehensively. Identifying key driver genes and understanding the condition-specific roles of different regulatory elements: visualization of regulatory networks, identification of essential driving genes, and comprehension of the context-dependent activities of various regulatory elements.

Finally, integrating transcriptomic information about geography with the temporal data emerging from time-course studies would add more dynamism to knowledge regarding gene regulation and cellular processes. Overcoming these issues and further pursuing such potential directions will keep multi-omics data analysis at the forefront in discovering Gene Regulation's intricate mechanisms, thus driving advances in personalized medicine, drug development, and understanding complex disorders.

4.5 Recent studies of network topology analysis

Such gene-gene interactions can improve the understanding of network topology analysis in a genetic architecture for complex traits and diseases, giving rise to more efficient diagnostic, prognostic, and therapeutic strategies. Network topology analysis refers to a method used to explain and analyze the layout of links within a biological network, usually concerned with gene regulation signal transduction routes or metabolic processes. Finally, it would be expected that genes interacting with large numbers of others are often key nodes of network topologies. These nodes can be crucial for the stability and functionality of genetic networks and identify hub genes, which are highly connected genes that may play critical roles in biological processes. Networks often have modular structures where clusters of genes interact more frequently among themselves than with other parts of the network, which can be crucial for understanding the functional organization of the genome. These modules can correspond to biological processes or pathways. Table 8 offers studies of recent investigations conducted within the field.

Graafland and Gutiérrez (2022) explores using Gaussian Bayesian Networks (GBNs) to reconstruct Gene Regulatory Networks from high-dimensional microarray data. The authors argue that GBNs, specifically the Hill Climbing (HC) algorithm, are well-suited for this task because they capture complex dependency structures present in GRN. The paper compares HC with popular UNM algorithms like Glasso and its variants (SFGlasso and

Table 8 Studies on network topology analysis

References	Contributions	Code availability
Devi and Singh (2020)	Proposed an approach for labeling graph vertices that works well across all network types	Not available
Graafland and Gutiérrez (2022)	Developed the Glasso algorithm, which prefers a diverse selection of edges between communities rather than a consistent selection of edges within communities	Not available
Monti et al. (2022)	Synthetic time series gene expression data were generated for various prototypical GRNs	https://github.com/jonathan-f/DA_RNN_GENEXP
Cui et al. (2022)	Introduced a neural network design with organized sparsity that learns gene representations, enhancing interaction detection's power	https://github.com/tycui/GWAS_NN
Jiang et al. (2022)	Used novel application of PMI in a biological context, demonstrating its potential for extracting meaningful biological relationships, identifying biomarkers, and developing more effective therapeutic strategies in cancer and other complex diseases	Not available

Bold indicates to emphasize the methodologies adopted in previous studies

HGlasso) commonly used for GRN reconstruction. UNMs encode relationships as undirected graphs, and GBNs use DAGs. According to the Bayesian Information Criterion, HC is a score-based algorithm that optimizes a score function to learn DAG structure. Glasso and variants depend on l_1 regularization to enforce sparsity on the network and often result in biased parameter estimation. There is no previous knowledge required by HC about the structure of the network—a virtue that makes it quite appropriate for diverse biological systems. HC prioritizes informative edges towards compact and interpretable networks; this network topology also allows for narrowing centrality measures to further candidate TFs and regulatory interactions.

Monti et al. (2022) present paper considers DA-RNNs in applications connected with time-series gene expression data for the prediction and analyzes the structure of gene regulatory networks. The authors develop the power of RNNs for temporal prediction with interpretability from the attention mechanism to derive insights into gene interactions. In DA-RNN, there is a dual-stage attention mechanism: input attention identifies which genes are relevant to the expression of the target gene. Temporal attention focuses on the most informative time points within those selected gene expressions. This allows the DA-RNN architecture to handle long-term dependencies and complex relationships in GRNs. This work uses a parallel DA-RNN scheme, wherein each gene is assigned its DA-RNN to make the prediction; hence, analysis of the whole network is possible. In this paper, the effect of noise on the prediction accuracy of the DA-RNN for different architectures of GRNs is investigated. In particular, the oscillating networks and those with external oscillatory signals have noise response patterns distinct from the other architectures. The mean squared error under the addition of noise to the models shows different levels of robustness across genes and network types. Hierarchical clustering and PCA on the MSE matrix further support the ability to differentiate GRN architectures based on noise response.

Devi and Singh (2020) the authors of this paper attempt to predict links when dealing with complex networks, looking to give a much more generalizable and automated approach than the previous methods. Topological features will be learned from the network

structure without relying on specific properties of the network or handcrafted features. Thus, a subgraph is extracted for each target link. This will involve exploring neighboring nodes until a predefined threshold size K , capturing the local network structure around the target link. The vertices' distance from the target link nodes in each subgraph is labeled by either the geometric mean or an arithmetic mean distance. This encodes the relative position and importance of each vertex within the subgraph. After labeling, the vertices are reordered, and then an adjacency matrix is formed, representing the subgraph structure and the relationships between vertices. The extracted features (adjacency matrices) are then used to train various machine learning models (e.g., SVM, kNN, Random Forest, Bagging) and a deep learning model (1D Convolutional Neural Network) to learn the topological patterns and predict the likelihood of the target link existing.

This paper Cui et al. (2022) introduced a novel deep-learning framework to identify gene–gene interactions affecting phenotypes and address the limitations of existing methods. Current approaches often focus on simple interactions between top SNPs (single nucleotide polymorphisms) within genes and assume specific interaction forms, leading to reduced power in detecting complex interactions.

Structured sparse neural network architecture learns gene representations from all SNPs within a gene, capturing complex relationships between genes and phenotypes. These scores quantify interactions between hidden nodes representing genes, measuring the interaction effect. A tailored permutation procedure assesses the significance of identified interactions, ensuring accurate identification of true interactions. It can detect complex interactions missed by traditional methods and learns complex relationships between genes and phenotypes without predefined. The authors presented in this paper a very promising deep learning framework for inferring complicated gene–gene interactions and gaining insights into the genetic basis of complex traits or diseases.

Jiang et al. (2022) suggested PMI, one of the measures of direct association, for analyzing gene coexpression networks in cancer transcriptome data. In particular, it tries to overcome some limitations of the traditional Pearson Correlation Coefficient and other correlation measures designed to study indirect associations, often producing redundant results. A measure called PMI will be used to calculate the strength of direct association between pairs of genes in transcriptome data. Modulation of this network is then realized through a thresholding approach, retaining only the most strongly significant PMI values within a network, making it sparser than PCC-based networks.

The edges of this PMI network are gene pairs that are compared with known gene-pair relationships emanating from several sources of omics data, such as transcriptional regulation, protein–protein interactions, and copy number variations. This procedure will allow the evaluation of the biological relevance and the specificity of the network using PMI compared to that constructed based on PCC.

These gene pairs, which have CNV or DNA methylation relationships identified in the PMI network, are used to classify patients into different groups based on their survival outcomes. Among these gene pairs, some that show a considerable prognosis value for survival analysis would be further shortlisted as biomarkers. PMI provides a more accurate and robust methodology to quantify the expression data of genes and further infer multi-omics data, especially in complex biological systems like cancer.

Trends, challenges and future directions in recent studies of network topology analysis
The three papers Monti et al. (2022), Graafland and Gutiérrez (2022), and Cui et al. (2022)

represent current trends and challenges of the inference of gene regulatory networks from expression data using different machine learning and statistical methods. Indeed, all three papers are based either on machine learning or deep learning techniques, setting out a growing reliance on these methods for GRN inference. This trend shows that the complexity of biological data increases and model expectations change—nonlinear relationships and higher-order interactions. Stuff like Monti et al. (2022) and Cui et al. (2022) remain more focused on gene–gene interaction, while Graafland and Gutiérrez (2022) examine which PNMs are witnesses to GRNs most generally. This highlights the diverse focuses within the field, ranging from specific interaction types to general network structures. Jiang et al. (2022) demonstrate using part mutual information (PMI) to integrate gene expression data with other omics data, such as copy number variation and DNA methylation. This trend underscores the importance of incorporating diverse data types to understand gene regulation comprehensively.

5 Evaluation of methods gene interaction prediction studies

We can understand recent progress-taking methods for several datasets by analyzing this state-of-the-art. Details about the approaches discussed in the work are summarized in Table 9.

The matrices are important in studying gene–gene interactions because they allow researchers to evaluate models, understand their behavior, and make informed judgments about future research and applications. Matrices such as AUC, AUROC, AUPR, F1, and MSE are important in gene–gene interaction research because they are used to assess the performance of models and algorithms. Here's a breakdown of how important each is: AUC (Area Under the ROC Curve) quantifies a model's ability to differentiate between two groups (for example, interacting versus non-interacting gene pairs). The interpretation of a higher AUC score suggests more discriminatory power. An AUC of 0.5 indicates random guessing, whereas an AUC of 1.0 represents flawless categorization. AUROC (Area Under the Receiver Operating Characteristic Curve) is virtually synonymous with AUC. The terms are often used interchangeably. AUPR (Area Under the Precision-Recall Curve) analyzes the balance between precision (the ratio of true positives to all predicted positives) and recall (the ratio of true positives to all actual positives). This is particularly useful in imbalanced datasets where one class predominates. The F1 score, the harmonic mean of precision and recall, offers a balanced measure of model performance. It considers both false positives and negatives, making it ideal for assessing models on balanced data sets. Meanwhile, MSE (Mean Squared Error) calculates the average of the squares of the differences between predicted and actual values, with lower MSE values signifying more accurate predictions.

Microarray (Al-Mashanji and Al-Rashi 2019) Convolutional Neural Network datasets typically involve thousands of genes but relatively few samples, demonstrating a challenge for accurate and robust GRN inference. High dimensionality and low sample size are addressed differently in each paper, with Monti et al. (2022) focusing on specific gene sets, Graafland and Gutiérrez (2022) employing sparsity constraints, and Jiang et al. (2022) utilizing PMI's ability to handle such data. Deep learning models, while powerful, can be difficult to interpret, hindering the understanding of the inferred GRNs. Cui et al. (2022) overcomes this challenge by utilizing Shapley interaction scores to quantify gene–gene interactions

Table 9 Summary of GRN prediction studies, including the type of prediction model, datasets used, and their performance evaluations

Study	Dataset	Metric	AUROC	AUPR
D3GRN (Chen et al. 2019)	DREAM4-Network1 DREAM5-Network1	—	—	0.175 0.373
GNE (Kc et al. 2019)	Yeast <i>E. coli</i>	—	0.710 0.653	0.683 0.658
Ma et al. (2020)	InSilico_Size10 InSilico_Size100 <i>Escherichia coli</i> Yeast	—	0.950 0.812 0.5522 0.963	0.854 0.280 0.0185 0.833
GENELink (Mao et al. 2023)	scRNA-seq (hHEP)	—	4.93	9.94
Shachaf et al. (2023)	Yeast 1-Size 50 <i>E. coli</i> 2-Size 100	—	—	—
(GRINCD) (Feng et al. 2023)	Silico data	—	0.027	0.010
GC-MERGE (Rothenberg 2019)	GM12878 K562 HUVEC	—	0.91 0.92 0.90	0.88
FGN (Kotiang and Eslami 2020)	<i>E. coli</i>	$p > 0.9$	—	—
DeepSEM (Shu et al. 2021)	(hHEP) (hESC) (mESC) (mDC)	—	1.0	—
Zito and Cutello (2023)	Silico-100 Sliico-10 SOS	—	0.55 0.55 0.61389	—
NCNN (Li et al. 2023)	GTE _x	—	—	—
GDPS (Jagodnik et al. 2023)	DisGeNET Human Protein Atlas ARCHS4	—	0.93	0.97
Mostavi et al. (2020)	Pan-cancer RNA-Seq data	ACC=93.9– 95.0%	—	—
Leng and Wu (2022)	TNBC	—	—	1
CLARIFY (Bafna et al. 2023)	Mouse visual Cortex, mouse hypothalamus	—	0.710817	1.48
Xue et al. (2023)	EDC	$p < 0.001$	—	—
Fofana et al. (2021)	Human ependymal microarray data	—	1.0	—
Wani and Raza (2021)	<i>E. coli</i> +TCGA	—	0.8985	0.6881
IntOMICS (Pačinková and Popovici 2022)	TCGA-COAD	—	1.0	—
BRANENet (Jagtap et al. 2022)	Yeast multi-omics	—	—	1.0
SMGR (Song et al. 2022)	ScRNA-seq data, scATAC-seq data	—	—	—
Devi and Singh (2020)	Yeast <i>E. coli</i>	$p > 0.9$	—	—
Wani and Raza (2019)	<i>E. coli</i> <i>S. cerevisiae</i> Homo sapiens	—	0.97 0.84 0.73	0.46 0.42 0.37

Table 9 (continued)

Study	Dataset	Metric	AUROC	AUPR
Graafland and Gutiérrez (2022)	Silico, <i>E. coli</i>	–	0.6	–
Monti et al. (2022)	(RNA-seq)	$P=0.024$ for RUNX1; $P=0.0076$ for FOS; $P=0.033$ for NFE2	–	–
Cui et al. (2022)	FINRISK	–	–	1.0

within the network. Disentangling direct and indirect relationships between genes remains a significant challenge. Jiang et al. (2022) address this by advocating for PMI, which focuses on direct associations, to refine the results of conventional indirect association measures. Combining the strengths of different methods, such as integrating deep learning with prior biological knowledge or utilizing causal inference techniques, could lead to more accurate and interpretable GRN inference. Moving beyond static GRNs and into models representing the temporal dynamics of gene regulation is critical for comprehending complicated biological processes and disease development. In this regard, researchers can aid in better understanding gene regulation and its involvement in a wide range of biological processes and disorders.

6 Discussion

The data memorized in Table 9 based on results from several gene regulatory network prediction studies shall be analyzed in this section. We want to extract lessons from these findings about the problems and gains made in this field. Both methods for the GRN inference are focused on the Area under the Precision-Recall Curve as their primary metric since both studies, D3GRN Chen et al. (2019) and Ma et al. (2020) used the same dataset from DREAM4. Let's analyze the results presented in both papers. Ma et al. (2020) reached an AUPR of 0.280, which is a pretty good but not exceptional performance; D3GRN from Chen et al. (2019) achieved a comparative AUPR of about 0.175. Compared with these models using the same train dataset, the performances of D3GRN are lower than those of Ma et al. (2020). The Ma et al. (2020) model focused on capturing complex, nonlinear gene interactions, making it more realistic than linear models. This was realized by concentrating on pairwise interactions that help sidestep the intricacies of including all possible genes to regulate and target every gene. The Ma et al. model of 2020 was compared with that of Kc et al.'s GNE model 2019 using *E. coli*.

It is, for example, reported that GNE performs well in differentiating between the true regulatory interactions from the false ones with an AUROC of 0.710 on both coli and yeast datasets, either by AUROC or AUPR. In contrast, Ma et al. have an AUROC of 0.963, showing excellent discriminatory power for the yeast dataset. Again, GNE is ahead of Ma et al. regarding AUPR with 0.658 against their variant's 0.0185. This further puts forward the superiority of the ability of GNE in prioritizing true interactions for *E. coli*, especially considering the huge imbalance between true and false interactions. Off the shelf, GNE explicitly models both topological and attribute proximity, which makes it possible to out-

line a more comprehensive understanding of gene relationships, hence improving the accuracy of predictions of interactions. This also discusses the impact of network sparsity and the importance of fine-tuning parameters for optimal performance. However, the advantages of the GNE model lie in the fact that it provides a powerful and efficient way for gene network inference by deep learning and integrating diverse biological data. RWRNET and the methodology have very good performance toward the inference of GRN but still show sensitivity to network size and evaluation metrics. For DREAM3, ten genes in both methods for both studies obtain high accuracy and TPR, which indicates their good performance on this small network.

Yu et al. reached a PPV of 1, indicating better precision in identifying true interactions. For DREAM3-50 genes and 100 genes, both methods maintain good accuracy, although the TPR values are lower because larger networks are more difficult to infer. RWRNET gave a bit higher TPR, while Yu et al. have a higher MCC, which suggests this method is better balanced in treating true and false predictions for the imbalanced case. In the SOS Network, RWRNET shows significantly higher TPR and PPV than Yu et al., indicating superior performance in identifying and prioritizing true interactions within this network. Both RWRNET and the method by Yu et al. demonstrate good performance on GRN inference tasks, but their strengths and weaknesses vary depending on the network and evaluation metric.

The model GENELink (Mao et al. 2023) focuses on seven scRNA-seq datasets of various cell types with diverse ground truth networks (STRING, non-specific ChIP-seq, cell-type-specific ChIP-seq, and LOF/GOF). Achieves significantly higher AUROC values compared to six baseline methods (GENELink, GNE, CNNC, DeepDRIM, GRN-Transformer, and GENIE3) across most datasets, indicating its effectiveness in discriminating true regulatory interactions from false ones and superior performance in terms of AUPR on a large majority of the benchmark datasets, suggesting its ability to prioritize true positive interactions while minimizing false positives accurately. The model (GRINCD) Feng et al. (2023), employs three datasets from the DREAM5 challenge (in silico, *E. coli*, and *S. cerevisiae*) and two scRNA-seq datasets (mDC and hESC) with different gold standards (STRING, non-specific ChIP-seq, and cell-type-specific ChIP-seq). AUROC and AUPR achieve superior or comparable performance to 17 baseline methods across various datasets and metrics, showcasing their effectiveness in discriminating and prioritizing true regulatory interactions. GNNLink's strength lies in its GCN-based interaction graph encoder and link prediction formulation, while GRINCD excels in integrating heterogeneous data and causal discovery using ANM. Each study's datasets' specific network structures and data characteristics likely influence their relative performance.

The combination of the novel CMIA inference in the study CMIA (Shachaf et al. 2023) algorithm with the KSG estimator consistently achieves the best or among the best median AUPR scores in most of the tested networks. The method's effectiveness in identifying true positive interactions while minimizing false positives. Running KSG with CMIA on the DREAM3 50-gene networks yielded a median improvement of 16% in AUPR compared to the gold standard. For the DREAM3 100-gene networks, the mean AUPR improvement for KSG with CMIA was 24%, while for the DREAM4 100-gene networks, the combination provided a mean improvement of 34%. In particular, progress in GRN inference performance, measured by AUPR, is significant for the KSG algorithm. Combining KSG with

the novel algorithm CMIA is a prospective approach toward achieving high-precision and recall-based reconstruction of GRNs.

The FGN paper by Kotiang and Eslami (2020) reports p values of greater than 0.90 for both SOS and AR networks when comparing predicted marginal probabilities of gene states against observed gene expression data. The strong positive linear relationship means that this model has properly captured the behavior of these networks under all sorts of experimental conditions. The papers GC-MERGE Rothenberg (2019), DeepSEM Shu et al. (2021), and GDPS Jagodnik et al. (2023) use AUROC for different but related tasks: predicting gene–disease associations, evaluating cell representations for cell-type identification, and classifying gene expression levels. The models employed in each study have distinct architectures: heterogeneous integrated graph model, beta-VAE-based deep generative model, and GCN model. Different data modalities were utilized, with HetIG-PreDiG incorporating gene expression and network information, DeepSEM utilizing scRNA-seq data, and GC-MERGE integrating histone modifications and Hi-C data. All papers leverage the advantages of AUROC, such as its insensitivity to class imbalance and its ability to provide a general measure of model performance across different classification thresholds.

Both papers Zito and Cutello (2023) and NCNN Li et al. (2023) demonstrate the potential of machine learning approaches for gene expression prediction and network inference. Zito and Cutello (2023) achieves low MSE values, indicating good gene expression prediction accuracy. The Reports AUC-ROC values surpass state-of-the-art SOS DNA Repair dataset methods, suggesting effective network inference. NCNN (Li et al. 2023) model reports lower MAE values compared to D-GEX and GCN, demonstrating improved prediction accuracy with NCNN focuses on prediction accuracy and does not explicitly evaluate network inference performance. Both papers Mostavi et al. (2020) and Leng and Wu (2022) present valuable contributions to gene expression analysis. While Paper 1 focuses on classifying and identifying cancer-related genes, Paper 2 delves into differential network inference and interaction-based analysis. Each approach offers unique insights into gene regulation and its role in various biological processes and disease contexts. Cancer type prediction by Mostavi et al. (2020) achieves high classification accuracy (93.9–95.0%) for distinguishing between 33 cancer types and normal samples, demonstrating the effectiveness of CNN models for cancer type prediction, and it shows robustness to noise, maintaining high accuracy even with significant levels of added noise in the data. Breast cancer subtype prediction achieves an average accuracy of 88.42% for predicting breast cancer subtypes using a simplified 1D-CNN model. Leng and Wu (2022) demonstrates the superior performance of CDN compared to other differential network inference methods in simulation studies, particularly for handling limited sample sizes and high dropout rates, and shows that CDN effectively leverages information from multiple sources (individuals/subtypes) to improve network inference accuracy.

CLARIFY Bafna et al. (2023) achieved a higher AUPRC ratio (1.48) compared to baseline methods Genie3 (1.40) and CeSpGRN (1.33). In inferring GRNs, even with its focus on joint refinement using AUROC and AUPRC ratios, provides quantitative evidence of CLARIFY's superior performance compared to existing methods, highlighting its potential for advancing the understanding of cell communication and gene regulation within tissues. The WGCNA model identified 15 co-expression modules, with the light-green module showing the strongest positive correlation with the PASI score ($r=0.9$, $p=4e-06$). WGCNA Application in Psoriasis Research demonstrates the utility of WGCNA for analyzing gene

expression data and identifying co-expression modules related to disease severity and treatment response in psoriasis. It combines clinical PASI scores with full-length transcriptome analysis to understand oxymatrine's effects better. It was demonstrated through a substantial decrease in PASI scores ($p < 0.001$) and improvement in skin lesions after oxymatrine treatment.

Both papers Fofana et al. (2021) and Pačínková and Popovici (2022) analyze gene expression data. The fact that both use AUC as a performance metric allows a quantitative comparison to show that their proposed methods have better average performance than existing approaches. Fofana et al. focus on incorporating assumptions and network information into single-omics (gene expression) analysis, while Pačínková and Popovici aim for multi-omics data integration and regulatory network inference. Both papers are Bayesian, though they are concerned with the methodology used. Fofana et al. applied hybrid p values and CAR, while Pačínková and Popovici applied an MCMC-based framework, including prior knowledge and estimating empirical knowledge. Fofana et al. primarily use simulated data for evaluation, while Pačínková and Popovici include both simulated and real datasets, providing additional biological context and validation.

BRANEnet Jagtap et al. (2022) outperforms baseline methods in TF-target prediction tasks, achieving higher AUPR scores with both average and weighted L2 edge feature operators. This suggests better accuracy in predicting regulatory relationships between TFs and their target genes. BRANEnet successfully learns features for differentially expressed biomolecules (genes, TFs, metabolites) involved in heat stress response in yeast and matrices; it plays a fundamental role in BRANEnet by representing network structure, capturing node similarities, and enabling efficient feature learning through matrix factorization. This approach contributes to the method's effectiveness, scalability, and versatility in integrating and analyzing multi-omics data for various downstream tasks. SMGR (Song et al. 2022) identify co-regulatory programs (genes with similar expression and chromatin accessibility) from joint scRNA-seq and scATAC-seq data. It performs SOMatic in identifying co-regulatory programs, achieving a higher CH index, Silhouette score, Dunn index, and ARI, while having a lower DB index, also outperforms SOMatic on real datasets, showing better cluster separation and compactness.

MKL-GRNI and iMTF-GRN, per (Wani and Raza 2019, 2021) respectively, attested to higher accuracy in inferring regulatory networks through conventional methods. However, when other data, including protein interactions, were considered, MKL-GRNI achieved better accuracy than iMTF-GRN. Each of the papers used experimental data that included *Homo sapiens* and *Escherichia coli* and demonstrated the effectiveness of the methods used. Concerning iMTF-GRN, it indicated a strong capacity to distinguish regulated and unregulated, achieving a high AUROC, which indicates how well the model handles data from multiple sources, increasing the capacity of the model to detect patterns. Regarding MKL-GRNI, the accuracy classification improves through parallel learning, demonstrates higher AUROC, and improves performance by integrating multiple data sources to detect important genes better. Based on the results achieved in the preceding benchmarking section of gene–gene association and gene–gene interaction forecasting, it is reasonable to summarize a global vision of the strengths and weaknesses of the considered GRN-based models. It would also be fair to assign that resolving the gaps and weaknesses present in the methods related to GRN inference, GGI prediction, and planning future experiments may be used

to inform subsequent iterations of GRN inference methods, GGI prediction methods, or experimental design. Therefore, it is necessary to evoke the contexts.

Cui et al. (2022) focus on genotype and phenotype data. The strengths of this paper include its ability to detect complex interactions and its ability to perform a well-calibrated permutation test. Weaknesses include computational costs for higher-order interactions, potential limitations in whole-genome analysis, and the need for further exploration of informative priors and effect sizes. Monti et al. (2022), Graafland and Gutiérrez (2022), and Devi and Singh (2020) all three papers deal with link prediction in complex networks. Monti et al. prioritize prediction accuracy and interpretability, Graafland and Gutiérrez focus on finding informative and parsimonious network structures, and Devi and Singh aim for a generalizable link prediction model.

7 Conclusions

This article provides a complete evaluation of GRN-based techniques for predicting gene–gene interactions. We selected 30 GRN-based models developed as of the date of our literature search in 2023. We evaluated these models using various criteria, including the nature of the prediction problem, the datasets used, and the features contained. Furthermore, we summarized each study’s strengths and shortcomings. In addition, we conducted a comparative assessment of GRN prediction performance in several research studies stratified by the datasets and methods used. This comprehensive study provides an overview of the present status of the discipline, shedding light on the advances and problems of prediction using machine learning approaches. The results obtained in the previous benchmarking section of gene–gene association and interaction prediction allow us to summarise the strengths and shortcomings of the evaluated GRN-based models. Naturally, addressing the shortcomings and limitations of these methods will likely inform and shape future development in GRN inference techniques, GGI prediction tools, and experiment planning. It is, therefore, essential to evoke these contexts.

Acknowledgements I would like to express my sincere thanks and appreciation to Al-Muthanna University, with which I am affiliated both professionally and academically, for its continued support and commitment to academic excellence. The support provided by [the College of Science at Al-Muthanna University/sci.college@mu.edu.iq] has been crucial in achieving the results presented here. Their dedication to fostering a rich and supportive research environment has played a major role in helping me achieve my academic and professional goals. suraibraheem@mu.edu.iq.

Author contributions Following the authorship policy provided by Springer, the contributions of each author to our manuscript titled “A Review of Methods for Gene Regulatory Networks and Analysis” are specified as follows: S.I. (Sura I. Mohammed Ali): Conceptualized the study, led the research design, and contributed to the writing of the manuscript, performed the systematic review, analyzed the data, and co-wrote the sections on computational methods, prepared all figures, and assisted in data visualization and interpretation. S.Z. (Sura Zaki Alrashid): Coordinated the manuscript preparation, edited the manuscript for critical intellectual content, and supervised the project.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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